

Yuxuan Zhang

(65) 8941-6785
e1216649@u.nus.edu

Homepage

GitHub

LinkedIn

Skills

Coding: Python, C/C++, Bash (Shell), SQL, MATLAB, R, Verilog, Java, C#, JavaScript/TypeScript
Tools: Linux, Git, CMake, Docker, CUDA, ROS/ROS2, PyTorch, TensorFlow, OpenCV, PCL, Open3D, Isaac Gym/Sim, MuJoCo, Unity, LabVIEW, Altium Designer, AutoCAD, SolidWorks, Multisim, Keil μ Vision, KiCad
Languages: Chinese (native), English (fluent)

Education

Master in Artificial Intelligence Systems

National University of Singapore

Aug 2025 – Present

Singapore

- GPA: TBD
- Coursework: Robotics, NLP, Pattern Recognition, ML

Graduate Diploma in System Analysis

National University of Singapore

Jul 2023 – Jan 2025

Singapore

- GPA: 4.11/5.00
- Coursework: ML, Full-stack Development, System Design

Bachelor in Intelligent Test & Control Engineering

Harbin Institute of Technology

Sep 2020 – Jun 2024

China

- GPA: 4.02/5.00
- Coursework: AI, Pattern Recognition & ML, Measurement & Control Systems, Embedded Systems, Distributed Systems, DSP

Publications

1. Chongkai Gao, Zixuan Liu, Zhenghao Chi, Junshan Huang, Xin Fei, Yiwen Hou, **Yuxuan Zhang**, Yudi Lin, Zhirui Fang, Zeyu Jiang, Lin Shao. “VLA-OS: Structuring and Dissecting Planning Representations and Paradigms in Vision-Language-Action Models.” *NeurIPS 2025*. [arXiv](#).
 - A unified VLA architecture series for controlled comparisons of task-planning paradigms and representations in robot manipulation.
 - Conducted controlled experiments across six manipulation benchmarks to analyze how planning representations affect performance across diverse settings.

Preprints

1. Chongkai Gao, Jieao Shi, Xuchuan Huang, Mu Zhaoyu, **Yuxuan Zhang**, et al. “ManiLadder: Benchmarking Manipulation Intelligence Frontier via a Categorized and Multi-Level Task Ladder.” *Target: CoRL 2026*.
 - A large-scale simulation benchmark with 112 tasks across four difficulty levels, spanning rigid, articulated, and deformable objects and multiple robot embodiments.
2. **Yuxuan Zhang**, et al. “CCTA: Safety- and Capability-Constrained Tool Arbitration for Long-Horizon Embodied Decision Making.” *Target: IEEE*.
 - A closed-loop tool-arbitration framework that combines capability filtering, VLM-conditioned tool scoring, and always-on safety projection.
3. Haziq Razali, **Yuxuan Zhang**, Qianli Xu, Yiannis Demiris. “Learning to Synthesize Novel Human-Object Interaction in Collaborative Task-Based Settings.” *AAAI 2026*.
 - A multi-view RGB-D framework for collaborative capture, reconstruction, and LLM-aided synthesis.

Selected Experience

Research Intern

Mar 2026 – Aug 2026

RAD @ AMD

Singapore

- Trained, fine-tuned, deployed, and profiled VLA policies (ACT, π_0 , SmolVLA) on AMD Strix Halo with Radeon 8050S.
- Developed Xilinx FPGA-based acceleration for VLA inference and low-latency robotic control.

AI Engineer Intern

Oct 2025 – Dec 2025

Fatfish AI

Singapore

- Integrated an NVIDIA Jetson-based inference system and MiniPC-based control stack for a humanoid robot with dexterous hands.
- Enabled LLM-based HRI, RGB-D perception, and closed-loop control for environment interaction and daily assistance.

Research Intern

Jun 2025 – Sep 2025

AdaComp Lab @ NUS

Singapore

- Developed LiDAR-based SLAM and route planning for autonomous quadruped navigation in dynamic environments.
- Implemented RGB-based object tracking to help visually impaired users interact with their surroundings.

Research Intern

May 2025 – May 2026

LinS Lab @ NUS

Singapore

- Studied VLA planning paradigms and manipulation benchmarking through VLA-OS and ManiLadder.

Research Intern

Mar 2025 – Dec 2025

*I2R @ A*STAR*

Singapore

- Developed multi-view RGB-D capture and reasoning pipelines for collaborative human-object interaction synthesis.

R&D Intern

Jul 2021 – Aug 2021

Changjiang Intelligent Control Co., Ltd.

China

- Developed multi-view stereo reconstruction, point-cloud processing, and robot-arm path planning for automated rust removal.

Tech Art Intern

Jan 2021 – Feb 2021

Boke City Co., Ltd. (Wuhan)

China

- Designed Unity-based VFX systems and implemented interactive logic in C#.

Selected Projects

Automated Calibration System for A/D Modules

Oct 2023 – May 2024

Undergraduate Thesis, Harbin Institute of Technology

China

- Built a 16-channel PCB with RS-232 firmware and integrated PXIe instrumentation with LabVIEW for automated calibration.

Automated Drone Docking & Charging Platform

Oct 2021 – Sep 2022

Innovation Project, Harbin Institute of Technology

China

- Developed an autonomous drone docking system using GPS and AprilTag localization with closed-loop landing control.

3D Port Reconstruction

Oct 2020 – Sep 2021

Project, Harbin Institute of Technology

China

- Reconstructed large-scale 3D port environments in Unity with VR interaction and dynamic simulation.